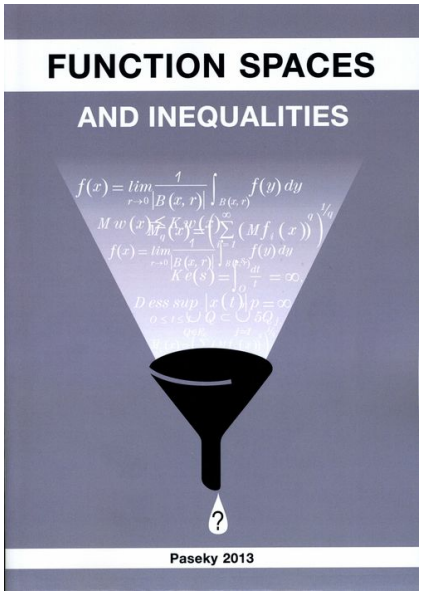
Forty Years of Muckenhoupt Weights

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Introduction

The full characterization of the weights w for which the Hardy-Littlewood maximal operator is bounded on $L^p(w)$ by means of the so-called A_p condition was achieved by B. Muckenhoupt and published in 1972 ([36]). Muckenhoupt's result became a landmark in the theory of weighted inequalities because most of the previously known results for classical operators had been obtained for special classes of weights (like power weights). The importance of the A_p classes became apparent soon after Muckenhoupt's work when it was shown that the weighted inequalities obtained for the Hardy-Littlewood operator were shared by some other significant operators of Fourier Analysis.

Since the A_p classes are related to the boundedness of several relevant operators, it will not be a surprise if we say that they have a rich structure. Several structural properties were obtained within about a decade after Muckenhoupt's result: characterization of A_1 weights, factorization and extrapolation, for instance. The literature on weights grew largely as shown by the long list of references of the survey [15] (1983). Already in 1981, J. Garnett's book [21] included a section on weights for the maximal operator and the Hilbert transform, and their relation to the space BMO. Shortly after that period appeared the book [20] by J. García-Cuerva and J. L. Rubio de Francia, the first book to have weighted inequalities as its main theme. From then on many textbooks incorporate the basic theory of weighted inequalities ([11], [24] and [46] appear in the references). The recently published book [7] contains an advanced treatment of weighted inequalities from the point of view of extrapolation. Its first chapter gives a historical account of the theory and is a rich source of bibliographic information.

A new ingredient in the theory of A_p weights was initiated with the thesis of S. Buckley, published in 1993 ([3]). He studied the growth of the norm of the Hardy-Littlewood maximal operator as a bounded operator on $L^p(w)$ in terms of the so-called A_p constant of w. It seems that the similar

question concerning other operators did not raise interest for several years. Nevertheless, after the answer given for the Beurling-Ahlfors transform by S. Petermichl and A. Volberg in 2002, it became one of the main themes in the theory of weights. New tools had to be developed, because the bounds provided by the usual techniques were not sharp. It has been only recently that the complete answer for the Calderón-Zygmund operators has been obtained ([29]). One can say now that the modern theory of weights pays special attention to the dependence on the constants of the weights.

The aim of this course is to present the classical theory of A_p weights proving the characterization of the weak-type and the strong-type inequalities for the Hardy-Littlewood maximal operator, and obtaining the sharp dependence on the A_p constants. The main properties of the A_p classes are also studied, considering in most cases the constants of the involved weights. We devote a chapter to the reverse Hölder inequality and another one to several characterizations of A_∞.

We introduce also maximal operators and weights defined from other bases. Sometimes the properties of the usual A_p classes extend immediately to this more general setting, but we also show with examples that there are well-known properties failing to hold for some bases.

One of the main features of the weighted inequalities is left outside this course: the study of the boundedness on $L^p(w)$ for $w \in A_p$ of other operators, mainly singular integral operators. Fortunately, this will be covered in this *Spring School on Analysis 2013* by the course of Carlos Pérez.

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Basic notation

- $L^p(\mu)$ is the space of functions whose p-th power is integrable with respect to μ; $\|\cdot\|_{p,\mu}$ is the norm; $\mu(A)$ is the measure of A with respect to μ.
- When μ is the Lebesgue measure we simply write L^p, $\|\cdot\|_p$, and $|A|$. In the same way, we do not mention explicitely the measure for integrals with respect to the Lebesgue measure.
- When $d\mu(x) = w(x)dx$, we write $L^p(w)$, $\|\cdot\|_{p,w}$, and $w(A)$.
- p' is the exponent conjugate to p: $\dfrac{1}{p} + \dfrac{1}{p'} = 1$ (with $\dfrac{1}{\infty} = 0$).
- χ_A is the characteristic function of the set A.
- For a cube Q, $l(Q)$ is its sidelength; kQ is the cube with the same center as Q and $l(kQ) = kl(Q)$.
- $B(x, r)$ is the ball centered at x with radius r.
- For a weight w, we write $\sigma = w^{1-p'}$, the value of p being given by the context.
- For a weight w and a cube Q, $w_Q = \dfrac{w(Q)}{|Q|}$ is the average of w over Q.
- When the number of a theorem, lemma or formula is of the form $A.j$ it refers to the appendix called *Toolbox* at the end of the paper.

Chapter 1

Maximal operator and A_p weights

1.1 Hardy-Littlewood maximal operator

We define the Hardy-Littlewood maximal operator for a locally integrable function f in $\mathbf{R}^n$ as

$$Mf(x) = \sup_{x \in Q} \frac{1}{|Q|} \int_Q |f|, \tag{1.1}$$

where the supremum is taken over the cubes[1] in $\mathbf{R}^n$ containing x, and can be infinity. If the cubes are taken centered at x, we have the *centered* maximal operator, which we will denote as M^c. Both versions are equivalent regarding the boundedness properties to be considered here, because we have the pointwise inequalities

$$M^c f(x) \leq Mf(x) \leq 2^n M^c f(x).$$

Another typical variant consists on using Euclidean balls instead of cubes. The corresponding centered and uncentered maximal functions are equivalent to those built from the cubes.

The Hardy-Littlewood maximal operator was introduced by G. H. Hardy and J. E. Littlewood in 1930 in the one-dimensional case (actually in $(0, +\infty)$) and extended to higher dimensions by N. Wiener in 1939 (see [27] and [47],

[1] The cubes considered here have sides parallel to the coordinate axes, although allowing cubes in all directions would only introduce a dimensional constant in the results.

respectively). The basic boundedness properties of M were already obtained in those papers:

- for $1 < p \leq \infty$, M is bounded on L^p, that is, $\|Mf\|_p \leq C_p \|f\|_p$;
- M is of weak-type $(1,1)$, that is,

$$\sup_{t>0} t|\{x \in \mathbf{R} : Mf(x) > t\}| \leq C\|f\|_1. \tag{1.2}$$

The boundedness in L^1 fails and the weak-type inequality (1.2) is the good substitute. It is easy to check that it fails globally, because $Mf(x) \geq C|x|^{-n}$ for big x. It can also fail locally: if $f(x) = x^{-1}(\log x)^{-2}\chi_{(0,1/2]}$ in $\mathbf{R}$, then Mf is not integrable near the origin.

The usual proof of the weak-type $(1,1)$ using covering lemmas gives also weighted inequalities. In this section we present the theorem and its proof without weights, with an improvement in the statement of the weak-type. The weighted version is in the next section.

Theorem 1.1. *(i) Let $f \in L^1(\mathbf{R}^n)$. There exists $C > 0$ depending only on n such that for all $t > 0$ it holds that*

$$|\{x : Mf(x) > t\}| \leq \frac{C}{t} \int_{\{x:|f(x)|>t/2\}} |f(x)|\, dx. \tag{1.3}$$

(ii) Let $1 < p \leq \infty$. There exists $C > 0$ depending only on n and p such that

$$\|Mf\|_p \leq C\|f\|_p.$$

Proof. Assume that f in a nonnegative function in L^1. Let $E_t = \{x : Mf(x) > t\}$. If $x \in E_t$, then there exists a cube Q containing x such that

$$\frac{1}{|Q|} \int_Q f(x)\, dx > t. \tag{1.4}$$

If K is a compact subset of E_t, we can cover K with a finite family of cubes satisfying (1.4). Applying Vitali's covering lemma (Lemma A.4) we extract the finite family of cubes $\{Q_j\}$ whose triples cover K. Then

$$|K| \leq \sum_{j=1}^{N} |3Q_j| \leq \sum_{j=1}^{N} \frac{3^n}{t} \int_{Q_j} f(x)\, dx \leq \frac{3^n}{t} \int_{E_t} f(x) dx.$$

Taking the supremum of $|K|$ over the compacts contained in E_t, we have the usual weak-type inequality (1.2). To obtain the improvement (1.3) we proceed as follows. Decompose $f = f_1 + f_2$, where $f_1(x) = f(x)$ if $f(x) \le t/2$, and $f_1(x) = 0$, otherwise. Since $Mf(x) \le Mf_1(x) + Mf_2(x)$, we have

$$\{x : Mf(x) > t\} \subset \{x : Mf_1(x) > t/2\} \cup \{x : Mf_2(x) > t/2\},$$

The first set of the right-hand side is empty. Apply the weak-type inequality to the second set to get (1.3).

To obtain part (ii) we represent the L^p-norm as in (A.1) and use (1.3):

$$\begin{aligned}\int_{\mathbf{R}^n} Mf(x)^p dx &= p\int_0^\infty t^{p-1}|\{x : Mf(x) > t\}|\,dt \\ &\le Cp\int_0^\infty t^{p-2}\int_{\{x:f(x)>t/2\}} f(x)dx \le C_p\int_{\mathbf{R}^n} f(x)^p dx.\end{aligned}$$

□

Remark 1.2. Part (ii) is usually obtained applying the Marcinkiewicz interpolation theorem A.6 with (1.2) and the trivial L^∞ estimate. Actually, (1.3) is the first step in the proof of the interpolation theorem.

A consequence of the weak-type inequality is the differentiation theorem (using Theorem A.9):

$$\lim_{r\to 0}\frac{1}{|Q(x,r)|}\int_{Q(x,r)} f(y)dy = f(x)$$

for almost all $x \in \mathbf{R}^n$. Here $Q(x,r)$ is the cube centered at x and sidelength $2r$. In particular, $f(x) \le Mf(x)$ a.e.

1.2 Fefferman-Stein inequality

In this section we show how to insert weights in the previous proof to obtain the weighted inequalities due to Fefferman and Stein ([17]).

Theorem 1.3. *Let u be a nonnegative measurable function. Then*

$$u(\{x : Mf(x) > t\}) \le \frac{C}{t}\int_{\mathbf{R}^n} |f(x)|Mu(x)\,dx. \tag{1.5}$$

Moreover, for $1 < p < \infty$ we have that

$$\|Mf\|_{p,u} \le C\|f\|_{p,Mu}.$$

Proof. We may assume that f is nonnegative and $f \in L^1(\mathbf{R}^n)$. (If $f \in L^1(Mw)$ is not in L^1, consider $f_n = f\chi_{B(0,n)}$, which is an increasing sequence of integrable functions pointwise convergent to f.) With the same notation as in the previous proof, take the selected cubes $\{Q_j\}$ covering the compact K to write

$$u(K) \le \sum_{j=1}^{N} u(3Q_j) \le \sum_{j=1}^{N} 3^n \frac{u(3Q_j)}{|3Q_j|} \frac{1}{t} \int_{Q_j} f(x)\,dx \le \frac{3^n}{t} \sum_{j=1}^{N} \int_{Q_j} f(x) Mu(x)\,dx.$$

This gives (1.5).

As in the previous theorem we can integrate only on $\{x : |f(x)| > t/2\}$ in the right-hand side of (1.5). This is enough deduce the strong-type inequality as before. □

The measures appearing in both sides of the inequalities of Theorem 1.3 are different, $u(x)dx$ and $Mu(x)dx$. They can be taken the same for those u satisfying the pointwise inequality $Mu(x) \le Cu(x)$ a.e. This condition will appear in the next section (Definition 1.5) with the name A_1.

1.3 Muckenhoupt weights

First we see that the measures for which a weak-type inequality holds are absolutely continuous with respect to the Lebesgue measure.

Theorem 1.4. *Let μ be a positive measure finite on compact sets. If the weak-type inequality*

$$\sup_{t>0} t\mu(\{x \in \mathbf{R}^n : Mf(x) > t\})^{1/p} \le C\|f\|_{p,\mu} \tag{1.6}$$

holds for some $1 \le p < \infty$, then μ is absolutely continuous with respect to the Lebesgue measure.

Proof. Let K be a compact set such that $|K| = 0$ and $\epsilon > 0$. There exists an open set V containing K such that $\mu(V \setminus K) < \epsilon$. Take $f = \chi_{V \setminus K}$.

Since K is of measure zero, $Mf(x) = 1$ for $x \in K$. The hypothesis (1.6) implies

$$t^p \mu(K) \le C\mu(V \setminus K) \le C\epsilon,$$

for all $t < 1$. Then $\mu(K) = 0$. □

In what follows, we consider only measures which are absolutely continuous with respect to the Lebesgue measure. Let us introduce the classes of weights involved in the main results.

Definition 1.5. (i) We say that a nonnegative locally integrable function w is in A_p for $1 < p < \infty$ if

$$\sup_Q \frac{w(Q)}{|Q|} \left(\frac{w^{1-p'}(Q)}{|Q|} \right)^{p-1} < +\infty, \tag{1.7}$$

where the supremum is taken over all the cubes Q in $\mathbf{R}^n$. The value of the supremum is the A_p constant of the weight w, and will be denoted by $[w]_{A_p}$.

(ii) We say that w is in A_1 if

$$Mw(x) \leq Cw(x) \quad \text{a.e.} \tag{1.8}$$

and $[w]_{A_1}$, the A_1 constant of w, is the smallest value of C for which the inequality holds.

Remark 1.6. The A_p condition takes a more symmetric form when stated in terms of $u = w^{1/p}$. It corresponds to

$$\sup_Q \frac{\|u\|_{L^p(Q)} \|u^{-1}\|_{L^{p'}(Q)}}{|Q|} < \infty.$$

Notation. For a weight w, we denote $\sigma = w^{1-p'}$ throughout the paper.

Muckenhoupt proved in [36] the following theorem.

Theorem 1.7. *Let $1 \leq p < \infty$. The operator M satisfies the weak-type inequality*

$$\sup_{t>0} t\, w(\{x : Mf(x) > t\})^{1/p} \leq C \|f\|_{p,w} \tag{1.9}$$

if and only if w is in A_p. Moreover, the constant C can be taken of the form $C_p [w]_{A_p}^{1/p}$, where C_p depends only on p and n.

Proof. Let Q be a cube, $S \subset Q$ with nonzero measure, and $f = \chi_S$. Then $Mf(x) \geq |S|/|Q|$ for $x \in Q$. If (1.9) holds, then

$$\left(\frac{|S|}{|Q|} \right)^p w(Q) \leq C w(S). \tag{1.10}$$

(Apply the weak-type inequality to $t < |S|/|Q|$ and let t tend to $|S|/|Q|$.)

If w is not zero almost everywhere, $w(S) = 0$ implies $|S| = 0$, so that $w(x) > 0$ a.e. On the other hand, w is locally integrable, otherwise $w(S) = \infty$ for every set S of positive measure.

Necessity for $p = 1$. Let $a = \inf\{w(x) : x \in Q\}$, where inf is the essential infimum. Then for each $\epsilon > 0$ there exists $S_\epsilon \subset Q$ such that $|S_\epsilon| > 0$ and $w(x) \le a + \epsilon$ for any $x \in S_\epsilon$. Inequality (1.10) for $p = 1$ and S_ϵ gives

$$\frac{w(Q)}{|Q|} \le C(a + \epsilon),$$

and so

$$\frac{w(Q)}{|Q|} \le C \inf_{x \in Q} w(x).$$

Therefore, for any cube Q,

$$\frac{w(Q)}{|Q|} \le Cw(x) \quad \text{a.e. } x \in Q. \tag{1.11}$$

This condition is equivalent to (1.8), with the same constant. Indeed, the set $\{x : Mw(x) > Cw(x)\}$ coincides with the union over the cubes with rational vertices of the subsets of such cubes for which (1.11) fails.

Sufficiency for $p = 1$. This is a consequence of Theorem 1.3.

Necessity for $p > 1$. Take $f = \sigma\chi_Q$. Then $Mf(x) \ge \sigma(Q)/|Q|$ for $x \in Q$. If (1.9) holds, then

$$\left(\frac{\sigma(Q)}{|Q|}\right)^p w(Q) \le C \int_Q \sigma^p w = C\sigma(Q), \tag{1.12}$$

which is the A_p condition (1.7)[2].

Sufficiency for $p > 1$. Proceeding as in the proof of Theorem 1.1 we select for a compact subset K of E_t the cubes $\{Q_j\}$ given by the covering lemma. Using Hölder's inequality we have

$$t < \frac{1}{|Q_j|} \int_{Q_j} f \le \frac{1}{|Q_j|} \left(\int_{Q_j} f^p w \right)^{1/p} \sigma(Q_j)^{1/p'}.$$

[2]Since we do not know a priori that σ is locally integrable it would be better to take $f = (w + \epsilon)^{1-p'}\chi_Q$. The bounds are independent of ϵ and the A_p condition follows. This condition implies that σ is locally integrable.

Then

$$t^p w(K) \le t^p \sum_j w(3Q_j) \le \sum_j w(3Q_j) \frac{3^{np}}{|3Q_j|^p} \left(\int_{Q_j} f^p w \right) \sigma(Q_j)^{p-1}$$
$$\le 3^{np} [w]_{A_p} \int_{E_t} f^p w.$$

Taking the supremum over the compacts contained in Q we get the result with the constant $3^n [w]_{A_p}^{1/p}$ in the right-hand side of (1.9). □

Using the Marcinkiewicz interpolation theorem with the trivial L^∞ estimate we obtain the following.

Corollary 1.8. *Let $1 \le q < p < \infty$ and $w \in A_q$. Then*

$$\|Mf\|_{p,w} \le C_{p,q} [w]_{A_q}^{\frac{1}{p}} \|f\|_{p,w}.$$

We get that M is bounded on $L^p(w)$ whenever $w \in \bigcup_{q<p} A_q$. We prove in Chapter 4 that this union coincides with A_p. This was the original way to proving that M is bounded on $L^p(w)$ for weights in A_p, but we give in Section 1.5 a different proof, which moreover provides sharp dependence on the constant of the weight.

Definition 1.9. We say that a measure μ has the *doubling property* if for some $C > 0$ and all cube Q it satisfies that

$$\mu(2Q) \le C\mu(Q).$$

The constant C is the doubling constant of μ.

The A_p weights have the doubling property. From (1.10) and the estimate of Theorem 1.7, it follows that there exists $C_{p,n}$ depending on p and n such that all w in A_p satisfies

$$w(2Q) \le C_{p,n} [w]_{A_p} w(Q).$$

Nevertheless, there exist weights with the doubling property but not in A_p for any $p \in (1, \infty)$. The first examples are due to C. Fefferman and B. Muckenhoupt ([16]).

Exercise. Check by direct computation that $|x|^\alpha \in A_p$ if and only if $-n < \alpha < n(p-1)$. Give an estimate of the A_p constant of $|x|^\alpha$.

1.4 Modified maximal operator

Given a positive measure μ, let us consider the centered maximal operator associated to μ given by

$$M_\mu^c f(x) = \sup_{r>0} \frac{1}{\mu(Q(x,r))} \int_{Q(x,r)} |f(y)|\, d\mu(y), \tag{1.13}$$

where $Q(x,r)$ is the cube centered at x and sidelength $2r$.

Theorem 1.10. *The operator M_μ^c satisfies the weak-type inequality*

$$\sup_{t>0} t\mu(\{x : M_\mu^c f(x) > t\}) \leq C_n \|f\|_{1,\mu}, \tag{1.14}$$

with a constant C_n depending on n but not on μ.

Moreover, for $1 < p < \infty$,

$$\|M_\mu^c f\|_{p,\mu} \leq C_{p,n} \|f\|_{p,\mu},$$

with $C_{p,n}$ depending on n and p, but not on μ.

Proof. The proof is very similar to the proof of Theorem 1.1, but this time we use Besicovitch covering lemma (Lemma A.5). We can do it because we have for each point in $\{x : M_\mu^c f(x) > t\}$ a cube centered at x for which the average appearing in the definition of (1.13) is greater than t. The constant appearing in (1.14) is the one given by Lemma A.5.

The strong-type inequality follows interpolating with the trivial L^∞ estimate. □

Remark 1.11. The noncentered version of (1.14) satisfies similar weak-type and strong-type inequalities if we are in one dimension. The weak-type $(1,1)$ is proved using Lemma A.3. In higher dimensions the boundedness can fail in general. Nevertheless, if the measure μ is doubling, the noncentered operator also satisfies weak-type and strong-type inequalities. This time we would use Lemma A.4. The constants in the inequalities depend on the doubling constant of the measure. Since A_p weights are doubling, this applies to the noncentered maximal operator associated to $d\mu(x) = w(x)dx$, where $w \in A_p$. But since the constants of the inequalities would depend on $[w]_{A_p}$ and we want to use the result to obtain (1.15), we need to go through the centered operator.

1.5 Weights and strong-type inequalities

The characterization of the strong-type inequalities for M is due to B. Muckenhoupt ([36]). We mentioned after Corollary 1.8 that it was possible to prove the boundedness of M on $L^p(w)$ for $w \in A_p$ if we were able to prove that every $w \in A_p$ is in some A_q for $q < p$. S. Buckley in his thesis proved the sharp growth of the operator norm of M in terms of the constant on the weight (published in [3]). Here we state and prove the theorem in Buckley's form, following a simple proof found by A. Lerner and published in [33].

Theorem 1.12. *Let $1 < p < \infty$ and $w \in A_p$. Then*

$$\|Mf\|_{p,w} \leq C_p [w]_{A_p}^{1/(p-1)} \|f\|_{p,w}, \tag{1.15}$$

where the constant C_p depends only on p and n.

Proof. Let f be a nonnegative function. Let Q be a fixed cube. Then

$$\frac{1}{|Q|}\int_Q f = \frac{\sigma(3Q)}{|Q|}\frac{1}{\sigma(3Q)}\int_Q f\sigma^{-1}\sigma \leq \frac{\sigma(3Q)}{|Q|} M_\sigma^c(f\sigma^{-1})(x),$$

for every $x \in Q$. We have used the fact that for every $x \in Q$ the smallest cube centered at x and containing Q is contained in $3Q$.

$$\begin{aligned}\left(\frac{1}{|Q|}\int_Q f\right)^{p-1} &\leq \left(\frac{\sigma(3Q)}{|Q|}\right)^{p-1}\frac{w(Q)}{|Q|}\left(\frac{1}{w(Q)}\int_Q M_\sigma^c(f\sigma^{-1})^{p-1}w^{-1}w\right)\\ &\leq 3^{np}[w]_{A_p}(M_w^c(M_\sigma^c(f\sigma^{-1})^{p-1}w^{-1}))(x_Q),\end{aligned}$$

where x_Q is the center of Q. Using that $Mf(x) \leq 2^n M^c f(x)$, we obtain

$$Mf(x) \leq 2^n 3^{np'}[w]_{A_p}^{1/(p-1)}(M_w^c(M_\sigma^c(f\sigma^{-1})^{p-1}w^{-1}))(x)^{1/(p-1)}.$$

Applying Theorem 1.10 to M_w^c on $L^{p'}(w)$ and to M_σ^c on $L^p(\sigma)$ we deduce the result. □

Remark 1.13. The strong-type inequality fails for $p = 1$. Indeed, if M is bounded on $L^1(w)$, then $w = 0$ a.e. If $w(x) > 0$ in a set of positive measure, by the differentiation theorem there exists x_0 such that $\lim_{r\to 0} w(Q_r)/|Q_r| = w(x_0)$, where Q_r is the cube centered at x_0 with sidelength r. Since $w(Q_r)$ and $|Q_r|$ are comparable, we can built a counterexample in $L^1(w)$ as in the unweighted case. For instance, in one dimension and $x_0 = 0$, the local counterexample mentioned in Section 1.1 works.

Exercise. Prove that if (1.15) holds with the constant $\Phi([w]_{A_p})$ at the right-hand side, then $\Phi(t) \geq ct^{1/(p-1)}$. In the one-dimensional case, take $f(x) = x^{s-1}\chi_{(0,1)}$ and $w(x) = |x|^{(p-1)(1-s)}$.

1.6 Maximal operators on bases

Let Ω be a measurable subset of $\mathbf{R}^n$. A basis $\mathcal{B}$ in Ω is a collection of measurable sets such that $0 < |B| < \infty$ for $B \in \mathcal{B}$ and $\Omega = \bigcup_{B\in\mathcal{B}} B$. We define on Ω the maximal operator associated with $\mathcal{B}$ as

$$M_{\mathcal{B}} f(x) = \sup_{B\in\mathcal{B}:x\in B} \frac{1}{|B|} \int_B |f(y)|dy.$$

With some additional assumption on $\mathcal{B}$ (for instance, $\mathcal{B}$ is countable or $\mathcal{B}$ is formed by open sets), the maximal operator applied to f is a measurable function.

Definition 1.14. We say that a nonnegative locally integrable function w is in $A_{p,\mathcal{B}}$ for $1 < p < \infty$ if

$$\sup_{B\in\mathcal{B}} \frac{w(B)}{|B|} \left(\frac{\sigma(B)}{|B|}\right)^{p-1} < +\infty.$$

We say that w is in $A_{1,\mathcal{B}}$ if

$$M_{\mathcal{B}} w(x) \leq Cw(x) \quad \text{a.e.}$$

The Hardy-Littlewood maximal operator used until now corresponds therefore to the basis of all open cubes in $\mathbf{R}^n$.

Definition 1.15. We say that $\mathcal{B}$ is a *Muckenhoupt basis* if for all $1 < p < \infty$, $M_{\mathcal{B}}$ is bounded on $L^p(w)$ for $w \in A_{p,\mathcal{B}}$.

1.6.1 Basis of dyadic cubes

An important variant of the maximal operator in $\mathbf{R}^n$ is obtained considering dyadic cubes. Rather than considering the typical covering of $\mathbf{R}^n$ (see Remark 1.17), in this work we will need the dyadic cubes obtained from a fixed cube. For convenience, we take half-open cubes, that is, translates of cubes of the form $[0, a)^n$.

Let us fix a (half-open) cube Q of sidelength l. First we divide each side of the cube into its two halves to obtain 2^n cubes of sidelength $l/2$. Let $\mathcal{Q}_1$ the family of these 2^n cubes. We proceed in a similar way with each one of the cubes of $\mathcal{Q}_1$ to obtain 2^{2n} cubes of sidelength $l/4$ forming the family $\mathcal{Q}_2$, and so on. The family $\mathcal{Q}_k$ is formed by 2^{kn} disjoint cubes of sidelength $2^{-k}l$, whose union is the original cube Q. All the cubes so obtained are the *dyadic cubes* (adapted to Q), and we denote as $\mathcal{D}$ the whole family of dyadic cubes. The associated maximal operator is the *dyadic maximal operator* that we denote as $M_{\mathcal{D}}$. We will not make explicit the dependence on Q in the notation.

By construction, it is easy to become convinced that each $x \in Q$ determines a unique sequence $\{Q_k(x)\}$ of dyadic cubes such that

$$x \in Q_k(x) \in \mathcal{Q}_k; \quad l(Q_k(x)) = 2^{-k}l : \quad Q_{k+1}(x) \subset Q_k(x).$$

Moreover, if $Q_k(x) = Q_k(y)$, then $Q_j(x) = Q_j(y)$ for every $j < k$.

Let f be in $L^1(Q)$ and nonnegative, and $t > f_Q$ (the average of f over Q). If x is a point such that $\mathcal{M}_{\mathcal{D}} f(x) > t$, there exists a unique k_0 such that for the cubes in the sequence $\{Q_k(x)\}$ we have

$$\frac{1}{|Q_{k_0}(x)|}\int_{Q_{k_0}(x)} f > t \quad \text{and} \quad \frac{1}{|Q_k(x)|}\int_{Q_k(x)} f \leq t \quad \text{for } k < k_0.$$

In other words, $Q_{k_0}(x)$ is the largest dyadic cube containing x over which the average of f is greater than t. Such maximal cubes for two different points in $E_t = \{x \in Q : M_{\mathcal{D}} f(x) > t\}$ are either the same or disjoint. It is then easy to check that we get a decomposition of E_t as a disjoint union of dyadic cubes, $\{Q_j\}$, with the following properties:

1. $t < \dfrac{1}{|Q_j|}\displaystyle\int_{Q_j} f \leq 2^n t.$

2. $|E_t| = \displaystyle\sum_j |Q_j| \leq \frac{1}{t}\int_{E_t} f;$

3. $f(x) \leq t$ for almost every $x \notin E_t$.

The first and second properties hold by construction of the sequence. From the weak-type inequality we obtain the differentiation theorem using Theorem A.9. As a consequence, $f(x) \leq M_{\mathcal{D}} f(x)$ a.e., and this implies the third

property. The decomposition of the cube Q obtained for each f and t is called the *Calderón-Zygmund decomposition*.

Using the second inequality of the first property and $\{x : f(x) > t\} \subset E_t$ we get a kind of "reverse" weak $(1,1)$ inequality for the maximal function:

$$|\{x \in Q : M_{\mathcal{D}} f(x) > t\}| \geq \frac{1}{2^n t} \int_{\{x \in Q : f(x) > t\}} f(x)\, dx. \tag{1.16}$$

Since $M_{\mathcal{D}} f(x) \leq Mf(x)$ pointwise, we can take M instead of $M_{\mathcal{D}}$ at the left-hand side of this inequaity.

Remark 1.16. A measure μ is dyadically doubling on Q if for every pair (Q', Q'') of dyadic cubes in Q, such that $Q' \subset Q''$ and $l(Q'') = 2l(Q')$, it holds that $\mu(Q'') \leq C\mu(Q')$ for an absolute constant (the *doubling constant* of μ). A doubling measure in $\mathbf{R}^n$ in the sense of Definition 1.9 is dyadically doubling on every cube Q.

Define the dyadic maximal operator associated to a dyadically doubling measure μ as

$$M_{\mathcal{D},\mu} f(x) = \sup_{k \geq 1} \frac{1}{\mu(Q_k(x))} \int_{Q_k(x)} |f(x)|\, d\mu(x),$$

where $\{Q_k(x)\}$ is the sequence of dyadic cubes associated to x. The Calderón-Zygmund decomposition and the "reverse" weak $(1,1)$ inequality (1.16) are then obtained with the measure μ replacing Lebesgue measure and the doubling constant of μ replacing 2^n. The details are left to the reader.

Remark 1.17. The basis of dyadic cubes in $\mathbf{R}^n$ is

$$\mathcal{D} = \{2^{-k}([0,1)^n + m) : k \in \mathbf{Z}, m \in \mathbf{Z}^n\}.$$

Let $\mathcal{Q}_k$ be the family of cubes for a fixed k. As before, each x is in just one of the cubes in $\mathcal{Q}_k$, and we get a (doubly infinite) sequence of cubes, $\{Q_k(x)\}$, with the same properties as before. There is a Calderón-Zygmund decomposition of $\mathbf{R}^n$ associated to each nonnegative integrable function f and any $t > 0$, again obtained choosing the largest dyadic cubes in which the average of f exceeds t. Since f is integrable, the averages of f tend to zero when the sidelength of the cubes tend to infinity, ensuring the existence of the searched maximal cubes. The inequality (1.16) is also valid in $\mathbf{R}^n$.

Although M is not pointwise controlled by (a multiple of) $M_{\mathcal{D}}$, it is still possible to get the weak-type $(1,1)$ of M from that of $\mathcal{M}_{\mathcal{D}}$, as can be seen in [11].

1.6.2 Basis of rectangles

Let $\mathcal{R}$ be the basis in $\mathbf{R}^n$ composed of the sets of the form $I_1 \times \cdots \times I_n$ where I_j is an open interval in $\mathbf{R}$ for $j = 1, \ldots, n$. The associated maximal operator is known as the *strong maximal operator*.

It is easy to check that the strong maximal operator is not of weak-type $(1,1)$: if f is the characteristic function of the unit ball, for large x the strong maximal operator behaves as $|x_1 \ldots x_n|^{-1}$. Nevertheless, it is of strong-type (p,p), a result which can be proved majorizing $M_{\mathcal{R}}$ by the composition of the one-dimensional maximal operators in each direction. Following the same idea and with some more elaboration it can be seen that it is a Muckenhoupt basis. See [11], [20] or [24].

It remains unknown to this date the value of the best exponent for the constant of the weight in the analogue of Theorem 1.12 for the strong maximal function.

1.6.3 Basis of rectangles in all orientations

Let now $\mathcal{B}$ contain the rectangles of the previous basis and all the rectangles obtained by rotating them in all directions. The corresponding maximal operator, sometimes called *universal maximal operator*, is unbounded on L^p for all finite p. It is easy to check that it is unbounded on L^p for $p \leq n$: take as f the characteristic function of the unit ball and realize that for large x the associated maximal operator behaves as $|x|^{-1}$. This is enough to ensure that $\mathcal{B}$ is not a Muckenhoupt basis. To prove that the universal maximal function is unbounded on L^p for all finite p, we use a Besicovitch-type set (see [24, Chapter 10] or [46, Chapter X]).

1.6.4 Basis of intervals $(0, b)$

Let $\mathcal{B}_0$ the basis of $(0, \infty)$ formed by all intervals of the form $(0, b)$, with $b > 0$. This basis has excellent covering properties because every collection of intervals in $\mathcal{B}_0$ is nested. We can proceed as we did for M to deduce that the associated maximal operator $M_{\mathcal{B}_0}$ is of weak-type (p,p) with respect to w if and only if $w \in A_{p,\mathcal{B}_0}$ for $1 \leq p < \infty$, and of strong-type (p,p) with respect to w if and only if $w \in A_{p,\mathcal{B}_0}$ for $1 < p < \infty$. Therefore $\mathcal{B}_0$ is a Muckenhoupt basis.

In [13] this basis and the associated $M_{\mathcal{B}_0}$ and $A_{p,\mathcal{B}_0}$ have been studied in relation with the *Calderón operator*, which is defined in $(0,\infty)$ as

$$Sf(t) = \frac{1}{t}\int_0^t f(x)dx + \int_t^\infty \frac{f(x)}{x}\,dx.$$

It is proved in [13] that the weights for $M_{\mathcal{B}_0}$ and S coincide. A remarkable property is that the class of weights for which both $M_{\mathcal{B}_0}$ and S are bounded on $L^1(w)$ is not trivial (that is, it contains weights that are not identically zero). More precisely, it is given by the weights satisfying $Sw(x) \leq Cw(x)$ a.e., a class strictly contained in $A_{1,\mathcal{B}_0}$.

Another difference of this basis with the usual basis of cubes is that we do not have here the analogue of Theorem 1.4 (see Remark 2.3 in [13]).

The basis of $\mathbf{R}^n$ formed by all the cubes (or Euclidean balls) centered at the origin has similar properties to those of $\mathcal{B}_0$.

1.6.5 Basis of Carleson cubes

The basis of Carleson cubes in $\mathbf{R}^{n+1}_+$ is given by

$$\mathcal{C} = \{Q \times (0, l(Q)) : Q \text{ cube in } \mathbf{R}^n\}.$$

It is possible to consider a *centered* version of the maximal operator considering for $x = (\overline{x}, x_{n+1})$ only cubes $Q \times (0, l(Q))$ with Q centered at $\overline{x}$, equivalent to the maximal operator $M_{\mathcal{C}}$ associated to the basis $\mathcal{C}$.

We leave as an exercise to check that $\mathcal{C}$ is a Muckenhoupt basis. The weights associated to $\mathcal{C}$ are those characterizing the weighted inequalities for the Bergman projection (see [1] and [2]). They are called *Bekollé-Bonami weights*.

Exercise. (a) Let $l > 0$ fixed and $Q(x,l)$ the cube centered at x with side-length $2l$. Define

$$T_l f(x) = \frac{1}{(2l)^n}\int_{Q(x,l)} |f|.$$

Prove that for $1 \leq p < \infty$ and all $w \in A_p$ it holds that

$$\|T_l f\|_{p,w} \leq [w]_{A_p}^{1/p}\|f\|_{p,w}.$$

(b) Let P_t be the Poisson kernel,

$$P_t(x) = c_n \frac{t}{(t^2 + |x|^2)^{\frac{n+1}{2}}},$$

where c_n is such that the integral of P_t is 1. Prove that for $1 \leq p < \infty$ and all $w \in A_p$ it holds that

$$\|P_t * f\|_{p,w} \leq C[w]_{A_p}^{1/p} \|f\|_{p,w}.$$

(c) Let $\mathcal{B}$ the basis of all cubes of side $2l$ (fixed). Prove that the associated $M_{\mathcal{B}}$ satisfies

$$\|M_{\mathcal{B}} f\|_{p,w} \leq C[w]_{A_p}^{1/p} \|f\|_{p,w},$$

for $1 \leq p < \infty$ and all $w \in A_p$.

Note that in all cases the exponent of the A_p constant is $1/p$, while for the Hardy-Littlewood maximal operator it is $1/(p-1)$.

Chapter 2

A_1 weights and factorization

2.1 First properties of the A_p weights

Several properties of the A_p weights are immediately deduced from the definition. Although they are true for any basis, we state them for the original A_p classes, for simplicity. The following theorem shows some of them with indication of the size of the constants of the involved weights.

Theorem 2.1. *Let $1 < p < \infty$.*

(i) *$w \in A_p$ if and only if $w^{1-p'} \in A_{p'}$ and $[w^{1-p'}]_{A_{p'}} = [w]_{A_p}^{p'-1}$.*

(ii) *If $q < p$, then $A_q \subset A_p$ and $[w]_{A_p} \leq [w]_{A_q}$.*

(iii) *If $w \in A_p$ and $0 \leq \alpha \leq 1$, then $w^\alpha \in A_p$ and $[w^\alpha]_{A_p} \leq [w]_{A_p}^\alpha$.*

(iv) *If $w_1, w_2 \in A_p$ and $0 \leq \alpha \leq 1$, then $w_1^\alpha w_2^{1-\alpha} \in A_p$ and*

$$[w_1^\alpha w_2^{1-\alpha}]_{A_p} \leq [w_1]_{A_p}^\alpha [w_2]_{A_p}^{1-\alpha}.$$

(v) *If $w_0, w_1 \in A_1$ then $w_0 w_1^{1-p} \in A_p$, and*

$$[w_0 w_1^{1-p}]_{A_p} \leq [w_0]_{A_1} [w_1]_{A_1}^{p-1}. \tag{2.1}$$

Property (i) is immediate from the definition. Properties (ii)-(iv) are consequences of Hölder's inequality. To prove the last property we use (1.11) in the form

$$w_j(x)^{-1} \leq [w_j]_{A_1} \frac{|Q|}{w_j(Q)} \quad \text{a.e. } x \in Q, \ j = 0, 1.$$

2.2 Building A_1 weights

Property (v) in the previous theorem gives a way to building A_p weights from A_1 weights. It seems then interesting to have a way to produce A_1 weights. We give two constructions: the first one is due to Coifman ([6]) and the second one to Rubio de Francia ([42] and [43]).

2.2.1 Coifman's construction

Theorem 2.2. *Let f be a locally integrable function such that $Mf(x) < +\infty$ a.e. and let $0 \leq \delta < 1$. Then $(Mf(x))^\delta \in A_1$ and $[(Mf(x))^\delta]_{A_1} \leq C(1-\delta)^{-1}$, where C depends only on the dimension, not on f or δ.*

Proof. We need to prove that there exists a constant C such that for every f, every cube Q and almost every $x \in Q$,

$$\frac{1}{|Q|}\int_Q Mf(y)^\delta dy \leq \frac{C}{1-\delta}\, Mf(x)^\delta.$$

Fix Q and decompose f as $f = f_1 + f_2$, where $f_1 = f\chi_{2Q}$. Then $Mf(y) \leq Mf_1(y) + Mf_2(y)$. Since $0 \leq \delta < 1$,

$$Mf(y)^\delta \leq Mf_1(y)^\delta + Mf_2(y)^\delta.$$

Since M is weak $(1,1)$, we have Kolmogorov's inequality of Lemma A.10. Then

$$\begin{aligned}\frac{1}{|Q|}\int_Q (Mf_1(y))^\delta dy &\leq \frac{A^\delta}{1-\delta}|Q|^{-\delta}\|f_1\|_1^\delta \leq \frac{A^\delta}{1-\delta}\left(\frac{1}{|Q|}\int_{2Q} f\right)^\delta \\ &\leq \frac{(2^n A)^\delta}{1-\delta} Mf(x)^\delta.\end{aligned}$$

On the other hand, for $x, y \in Q$ we have $Mf_2(y) \leq CMf_2(x)$, with a constant depending only on the dimension. Indeed, if Q' is a cube containing y and intersecting $(2Q)^c$, then $2Q'$ contains x. □

The construction of A_1 weights can be extended to some other maximal operators associated to bases, but not to all of them. For instance, if we consider the strong maximal operator associated to the basis of rectangles in $\mathbf{R}^n$ with sides parallel to the coordinate axes (Section 1.6.2), the corresponding statement is false (see [45]).

2.2.2 Rubio de Francia algorithm

J. L. Rubio de Francia proposed a construction of A_1 weights which can be carried out in very general situations. This construction has been called *Rubio de Francia algorithm*. We state the result in a more general way than the building of A_1 weights.

Theorem 2.3. *Let T be a nonnegative sublinear operator*[1] *bounded on $L^p(\mu)$ with norm $\|T\|_{p,\mu}$, where $p \geq 1$ and μ is a positive measure on some measurable space. Given $u \in L^p(\mu)$, nonnegative, define*

$$Ru(x) = \sum_{k=0}^{\infty} \frac{T^k u(x)}{(2\|T\|_{p,\mu})^k},$$

where T^0 is the identity and $T^k u = T(T^{k-1}u)$. Then

(i) $u(x) \leq Ru(x)$ *a.e.;*

(ii) $\|Ru\|_{p,\mu} \leq 2\|u\|_{p,\mu}$*;*

(iii) $T(Ru)(x) \leq 2\|T\|_{p,\mu} Ru(x)$ *for almost every x.*

The series defining Ru clearly converges in the norm of $L^p(\mu)$ and the first two properties are easy to check. To obtain (iii), we apply the property

$$T\left(\sum_{k=0}^{\infty} u_k\right) \leq \sum_{k=0}^{\infty} Tu_k \quad \text{a.e.}$$

The third property implies that we are building an "A_1 weight for T". If we choose T to be M, $1 < p < \infty$, and $w \in A_p$, for each $u \in L^p(w)$ we built an A_1 weight majorizing u pointwise, but with comparable norm in $L^p(w)$.

2.3 Factorization

The converse of property (v) of Theorem 2.1 also holds and it is called *factorization property*[2]. This was conjectured by B. Muckenhoupt and proved by P. Jones ([32]). The original proof was quite difficult and was subsequently simplified using the algorithm of the previous theorem. Rubio de Francia's approach used vector-valued inequalities and was further simplified in [5], where the first constructive proof appeared.

[1] See Section A.4 for the definition of sublinear operator.

[2] Property (v) is called *reverse factorization* in [7].

Theorem 2.4. *Let $w \in A_p$, $1 < p < \infty$. There exist w_0 and w_1 in A_1 such that $w = w_0 w_1^{1-p}$. Moreover, w_0 and w_1 can be taken such that*

$$[w]_{A_p} \le [w_0]_{A_1} [w_1]_{A_1}^{p-1} \le C[w]_{A_p}^2. \tag{2.2}$$

Proof. Write $u = w_1^{p-1}$, so that $w_1 = u^{p'-1}$ and $w_0 = wu$. We want u such that

$$M(wu) \le Cwu \qquad \text{and} \qquad M(u^{p'-1}) \le Cu^{p'-1} \qquad \text{a.e.},$$

that is,

$$w^{-1} M(wu) \le Cu \qquad \text{and} \qquad M(u^{p'-1})^{p-1} \le Cu \qquad \text{a.e.}$$

Take the operator

$$Tf = w^{-1} M(wf) + M(f^{p'-1})^{p-1}.$$

Since T is sublinear, we can apply the Rubio de Francia algorithm to built a weight as desired if we can find a space in which it is bounded. Since $w \in A_p$, it is easy to check that T is bounded on $L^{p'}(w)$ using the weighted inequalities of M and property (i) of the A_p weights in Theorem 2.1. Building Rf as in Theorem 2.3 for any $f \in L^{p'}(w)$, and choosing $u = Rf$ we have the needed w_0 and w_1.

Using the quantitative version of the weighted inequalities (Theorem 1.12) we can check that $\|T\|_{p',w} \le C[w]_{A_p}$. From this and property (iii) of Theorem 2.3 we have

$$[w_0]_{A_1} \le C[w]_{A_p} \quad \text{and} \quad [w_1]_{A_1} \le C[w]_{A_p}^{1/(p-1)}.$$

This gives the second inequality in (2.2). □

The same proof shows that Theorem 2.4 is valid for weights associated to general Muckenhoupt bases, except the second inequality in (2.2), which depends on the estimate (1.15) for the norm of M on $L^p(w)$.

We state in the next proposition other forms of reverse factorization which will be used in the next chapter.

Proposition 2.5. (a) *Let $1 \le p < p_0 < \infty$. If $w \in A_p$ and $u \in A_1$, then wu^{p-p_0} is in A_{p_0} and*

$$[wu^{p-p_0}]_{A_{p_0}} \le [w]_{A_p} [u]_{A_1}^{p_0-p}.$$

(b) *Let* $1 < p_0 < p < \infty$. *If* $w \in A_p$ *and* $u \in A_1$, *then* $(w^{p_0-1}u^{p-p_0})^{1/(p-1)}$ *is in* A_{p_0} *and*

$$[(w^{p_0-1}u^{p-p_0})^{1/(p-1)}]_{A_{p_0}} \leq [w]_{A_p}^{\frac{p_0-1}{p-1}} [u]_{A_1}^{\frac{p-p_0}{p-1}}.$$

They are obtained using the definition of A_p, property (1.11) of A_1 weights and Hölder's inequality, and we leave the proofs to the reader. Actually, they are dual of each other and the second one can be obtained from the first one and the property (i) of Theorem 2.1. The result in Proposition 2.5 is true also for weights associated to general bases.

The converse factorizations also hold, and they are obtained from Theorem 2.4. For instance, if $1 \leq p < p_0 < \infty$ and $w \in A_{p_0}$, there exist $u \in A_1$ and $v \in A_p$ such that $w = vu^{p-p_0}$. Indeed, write $w = w_0 w_1^{1-p_0}$ with $w_0, w_1 \in A_1$ using Theorem 2.4, and take $v = w_0 w_1^{1-p}$ and $u = w_1$.

Chapter 3

Extrapolation

3.1 First extrapolation theorem

Using the factorization of A_p weights and Riesz-Thorin interpolation (Theorem A.7) we can easily prove the following theorem.

Theorem 3.1. *Let $1 \leq p_0 < p_1 < \infty$. Assume that T is a linearizable operator*[1] *bounded on $L^{p_0}(w)$ for all $w \in A_{p_0}$ and on $L^{p_1}(w)$ for all $w \in A_{p_1}$. Then T is bounded on $L^p(w)$ for all $w \in A_p$.*

Proof. Let $w \in A_p$. Factorize it as $w = w_0 w_1^{1-p}$. From the assumptions, the operator mapping f to $w_1^{-1} T(w_1 f)$ is bounded on $L^{p_0}(w_0 w_1)$ and on $L^{p_1}(w_0 w_1)$. The result follows using Riesz-Thorin interpolation. □

The surprising discovery of J. L. Rubio de Francia was that actually one of the assumptions of the previous theorem is enough to get the same conclusion, so we pass from an "interpolation" setting to an "extrapolation" one. The theorem can be stated in the following form: *if for some $p_0 \in [1, \infty)$ the operator T is bounded on $L^{p_0}(w)$ for all $w \in A_{p_0}$, with norm bounded by a constant depending only on the A_{p_0} constant of the weight, then T is bounded on $L^p(w)$ for all $w \in A_p$ and all $p \in (1, \infty)$.*

The original statement of the extrapolation theorem by Rubio de Francia was given for sublinear operators and proved through an equivalence between vector-valued inequalities and weighted norm inequalities. As for the factorization theorem, constructive proofs were given soon afterwards. The first constructive proof was published by J. García-Cuerva in [19].

[1] The definition of linearizable operator is in Remark A.8.

There are several variants and extensions of the extrapolation theorem. Even the original version was more general that the one we stated above. One of the most important changes in our view of the extrapolation theorem is that it can be stated for pairs of functions, free of operators. It seems that this observation appears in [9, Remark 1.11] for the first time. Working with pairs of functions offers several advantages as can be seen, for instance, in Sections 3.2 and 3.3 of [7]. Since it does not add any difficulty to the proof of the theorem, this will be the version in our statements.

The extrapolation theorem has been applied in recent years to the behaviour with respect to the constant of the weight of the norm of some operators on $L^p(w)$ spaces. To be used in such situation we need a quantitative version of the extrapolation: assuming that the bound in the hypothesis is given as a function of $[w]_{A_{p_0}}$, we obtain in the conclusion a bound which is a function of $[w]_{A_p}$. The optimal version was obtained in [10]. The proof we give here is from [12]. The reader who is not interested in the behaviour of the constants can avoid the details concerning such dependence.

Theorem 3.2. *Assume that for some family of pairs of nonnegative functions, (f,g), for some $p_0 \in [1,\infty)$, and for all $w \in A_{p_0}$ we have*

$$\|g\|_{p_0,w} \leq \Phi([w]_{A_{p_0}})\|f\|_{p_0,w}, \tag{3.1}$$

where Φ is an increasing function defined on $[1,+\infty)$. Then for all $1<p<\infty$ and all $w \in A_p$ we have

$$\|g\|_{p,w} \leq C_1\Phi(C_2[w]_{A_p}^{\max(1,\frac{p_0-1}{p-1})})\|f\|_{p,w}. \tag{3.2}$$

Proof. Case $p<p_0$. Let $f \in L^p(w)$ for $w \in A_p$. Built Rf using the Rubio de Francia algorithm of Theorem 2.3. From the first reverse factorization in Proposition 2.5 we deduce that $w(Rf)^{p-p_0} \in A_{p_0}$. Then

$$\begin{aligned}
\|g\|_{p,w} &= \left(\int_{\mathbf{R}^n} g^p\, w\, (Rf)^{\frac{p}{p_0}(p-p_0)}(Rf)^{\frac{p}{p_0}(p_0-p)}\right)^{\frac{1}{p}} \\
&\leq \left(\int_{\mathbf{R}^n} g^{p_0} w (Rf)^{p-p_0}\right)^{\frac{1}{p_0}} \|Rf\|_{p,w}^{1-\frac{p}{p_0}} \\
&\leq \Phi([w(Rf)^{p-p_0}]_{A_{p_0}}) \left(\int_{\mathbf{R}^n} f^{p_0} w (Rf)^{p-p_0}\right)^{\frac{1}{p_0}} 2^{1-\frac{p}{p_0}}\|f\|_{p,w}^{1-\frac{p}{p_0}},
\end{aligned}$$

where we applied respectively Hölder's inequality, assumption (3.1) with the weight $w(Rf)^{p-p_0}$, and property (ii) of Theorem 2.3. Property (iii) of the same theorem together with Theorem 1.12 says that

$$[Rf]_{A_1} \leq 2\|M\|_{p,w} \leq C[w]_{A_p}^{1/(p-1)}.$$

Thus

$$[w(Rf)^{p-p_0}]_{A_{p_0}} \leq [w]_{A_p}[Rf]_{A_1}^{p_0-p} \leq C_2[w]_{A_p}^{(p_0-1)/(p-1)}.$$

On the other hand, $(Rf)^{p-p_0} \leq f^{p-p_0}$ using property (i) of Theorem 2.3 and the fact that the exponent is negative.

Case $p > p_0$. Using duality we have

$$\left(\int_{\mathbf{R}^n} g^p w\right)^{\frac{p_0}{p}} = \sup\{\int_{\mathbf{R}^n} g^{p_0} hw : h \geq 0 \text{ and } \int_{\mathbf{R}^n} h^{\frac{p}{p-p_0}} w = 1\}.$$

For such an h let us define H by $H^{p'}w^{1-p'} = h^{\frac{p}{p-p_0}}w$, that is, $h = (Hw^{-1})^{\frac{p-p_0}{p-1}}$. Then $H \in L^{p'}(w^{1-p'})$ with norm 1. Built RH using the Rubio de Francia algorithm. From the second reverse factorization in Proposition 2.5 it follows that $w^{\frac{p_0-1}{p-1}}(RH)^{\frac{p-p_0}{p-1}} \in A_{p_0}$. Then

$$\begin{aligned}\int_{\mathbf{R}^n} g^{p_0} hw &= \int_{\mathbf{R}^n} g^{p_0} w^{\frac{p_0-1}{p-1}} H^{\frac{p-p_0}{p-1}} \leq \int_{\mathbf{R}^n} g^{p_0} w^{\frac{p_0-1}{p-1}} (RH)^{\frac{p-p_0}{p-1}} \\ &\leq \Phi([w^{\frac{p_0-1}{p-1}}(RH)^{\frac{p-p_0}{p-1}}]_{A_{p_0}})^{p_0} \int_{\mathbf{R}^n} f^{p_0} w^{\frac{p_0-1}{p-1}} (RH)^{\frac{p-p_0}{p-1}},\end{aligned}$$

where we have used property (i) of the Rubio de Francia algorithm in the first inequality, and the hypothesis (3.1) with the weight $w^{\frac{p_0-1}{p-1}}(RH)^{\frac{p-p_0}{p-1}}$ in the second one. Since

$$[RH]_{A_1} \leq 2\|M\|_{p',w^{1-p'}} \leq C[w^{1-p'}]_{A_p'}^{1/(p'-1)} = C[w]_{A_p},$$

we obtain

$$[w^{\frac{p_0-1}{p-1}}(RH)^{\frac{p-p_0}{p-1}}]_{A_{p_0}} \leq [w]_{A_p}^{\frac{p_0-1}{p-1}}[RH]_{A_1}^{\frac{p-p_0}{p-1}} \leq C'[w]_{A_p}.$$

To conclude use Hölder's inequality and property (ii) of Theorem 2.3. □

Remark 3.3. Disregarding the dependence on the constants, the proof can be written with some modifications using the A_1 weights given by Coifman's construction in Theorem 2.2. But we need the openness property that will be obtained later (Theorem 4.5). For instance, in the case $p < p_0$ instead of $w(Rf)^{p-p_0}$ we would use $w(Mf)^{p-p_0}$. This can be written as $w[(Mf)^{(p_0-p)/(p_0-q)}]^{q-p_0}$, which is in A_{p_0} since $w \in A_q$ and $(Mf)^{(p_0-p)/(p_0-q)} \in A_1$.

The proof provided in [11] follows a similar idea but limits the first part of the proof ($p < p_0$) to the A_1 weights. The order of the exposition is crucial, because the result for all values of p is obtained from the A_1 weights. The openness property is again necessary.

Although these approaches work for the usual A_p classes, they cannot be applied to weights in other Muckenhoupt bases. There are several reasons for this: (i) Coifman's construction might fail; (ii) even if $(M_{\mathcal{B}}f)^\delta$ happens to be in $A_{1,\mathcal{B}}$, the inequality $f \le M_{\mathcal{B}}f$ might fail (even with a constant); (iii) the openness property need not to hold as we will see in the next chapter.

The proof given above, using the Rubio de Francia algorithm, is valid for any Muckenhoupt basis, except for the function of $[w]_{A_p}$ obtained in the conclusion. Indeed, we used the estimate given in Theorem 1.12, which can be different for other bases.

It is customary to split the proof into two parts but the reader will find in [7] a unified proof avoiding such splitting.

Remark 3.4. The exponent of $[w]_{A_p}$ appearing in (3.2) is optimal. There are operators, like the Hilbert transform, for which the inequality

$$\|Hf\|_{p,w} \le C[w]_{A_p}^{\max(1,\frac{1}{p-1})}\|f\|_{p,w}$$

is sharp, in the sense that it does not hold with a smaller exponent. It we take (3.1) for $p_0 = 2$ and $\Phi(t) = Ct$, the conclusion gives the sharp exponent for other values of p.

Actually, a similar inequality holds for Calderón-Zygmund operators. Before it was proved by T. Hyönen ([29]), the problem was known as A_2 *conjecture*. The reason of this name is that it was enough to obtain the bound in $L^2(w)$ with linear growth in $[w]_{A_2}$ to get the full result for all values of p by extrapolation. The introduction of [29] gives information about the history of the search for sharp bounds for several operators.

3.2 Variants of the extrapolation theorem

We present in this section other extrapolation theorems, stated without expliciting the dependence on the constants of the involved weights. We invite the interested reader to complete the details concerning the constants.

An apparently more general form of extrapolation is obtained by assuming that the index of the class of weights and the exponent of the Lebesgue space are different. The statement with pairs of functions makes it a simple corollary to the previous theorem.

Corollary 3.5. *Assume that for some $1 \leq s_0 < \infty$, some $p_0 > 0$ and all $w \in A_{s_0}$ we have*

$$\|g\|_{p_0,w} \leq C\|f\|_{p_0,w},$$

for a family of pairs of nonnegative functions (f, g). Then for $p > p_0/s_0$ and $s = ps_0/p_0$,

$$\|g\|_{p,w} \leq C\|f\|_{p,w},$$

for all $w \in A_s$.

Write the assumption as

$$\|g^{p_0/s_0}\|_{s_0,w} \leq C\|f^{p_0/s_0}\|_{s_0,w},$$

and apply Theorem 3.2 to the pairs $(f^{p_0/s_0}, g^{p_0/s_0})$ to prove this corollary.

Note that in the statement of Corollary 3.5 we are not assuming $p \geq 1$. But we are not really working with quasi-norms because when written in terms of $(f^{p_0/s_0}, g^{p_0/s_0})$, we get normed Lebesgue spaces.

Another direct consequence of the extrapolation theorem for pairs is the fact that the assumption can be limited to weak-type inequalities.

Corollary 3.6. *Assume that for $p_0 \geq 1$ and all $w \in A_{p_0}$ we have the weighted weak-type inequality*

$$\sup_{t>0} tw(\{x : g(x) > t\})^{1/p_0} \leq C\|f\|_{p_0,w}, \tag{3.3}$$

for a family of pairs of nonnegative functions (f, g), with constant depending only on the A_{p_0} constant of w. Then for $1 < p < \infty$ we have

$$\sup_{t>0} tw(\{x : g(x) > t\})^{1/p} \leq C\|f\|_{p,w},$$

for all $w \in A_p$.

Let $E_t = \{x : g(x) > t\}$. Write (3.3) as

$$\|t\chi_{E_t}\|_{p_0,w} \leq C\|f\|_{p_0,w},$$

and apply Theorem 3.2.

When applied to a sublinear operator (that is, when $g = Tf$ with T sublinear), using the openness property of the A_p weights of Theorem 4.5 and Marcinkiewicz interpolation theorem (Theorem A.6) the conclusion of the theorem can be extended to strong-type inequalities.

A definitely more general version of extrapolation is obtained if the exponents on both sides of the inequalities are permitted to be different (*off-diagonal extrapolation*). Thus we have the following theorem.

Theorem 3.7. *Let* $1 \leq p_0 < \infty$ *and* $0 < q_0, r_0 < \infty$. *Let* $\gamma = \dfrac{1}{r_0} + \dfrac{1}{p_0'}$. *Assume that for* $w^{r_0} \in A_{r_0\gamma}$ *it holds that*

$$\|g\|_{q_0,w^{q_0}} \leq C\|f\|_{p_0,w^{p_0}}.$$

Then for all $1 < p < \infty$ *and* $0 < q, r < \infty$, *such that*

$$\frac{1}{q} - \frac{1}{q_0} = \frac{1}{r} - \frac{1}{r_0} = \frac{1}{p} - \frac{1}{p_0},$$

and all $w^r \in A_{r\gamma}$ *we have*

$$\|g\|_{q,w^q} \leq C\|f\|_{p,w^p}.$$

Although more involved, the proof can be done in a similar way to that of Theorem 3.2. We invite the reader to provide the details. Otherwise, they can be found in [12].

The case $q_0 = r_0 > p_0$ is due to E. Harboure, R. Macías and C. Segovia for the $A_{p,q}$ classes of Muckenhoupt and Wheeden, related to the boundedness of the fractional integral operator I_α from $L^p(w^p)$ to $L^q(w^q)$ when $1/q - 1/p = \alpha/n$ (see [26]). It seems that the general case as given in the statement had not been considered before appearing in [12], but the proof can be carried out without further restrictions.

For more results concerning extrapolation theorems the reader is referred to [7] and [12].

Chapter 4

Reverse Hölder inequalities

4.1 Reverse Hölder inequalities for A_p weights

Reverse Hölder inequality is a characteristic property of A_p weights. It played a crucial role in the original proofs of the characterization of the strong-type weighted inequalities for M, which were obtained from Corollary 1.8 using the Marcinkiewicz interpolation theorem, together with the openness property of the A_p classes (see Theorem 4.5 below).

Several proofs appeared later avoiding the use of the reverse Hölder inequality to get the strong type. This is the case for the one we gave in Chapter 1 (Theorem 1.12). Even if it is not needed for such purpose, the reverse Hölder inequality is important enough as to deserve special attention.

Theorem 4.1. *Let $w \in A_p$, $1 \leq p < \infty$. Then there exist constants $C > 0$ and $r > 1$, depending only on p and the A_p constant of w, such that for any cube Q,*

$$\left(\frac{1}{|Q|}\int_Q w^r\right)^{\frac{1}{r}} \leq \frac{C}{|Q|}\int_Q w. \tag{4.1}$$

The converse with constant 1 is Hölder's inequality. This is the reason of the name given to inequality (4.1).

Before giving the proof of Theorem 4.1 in full generality we present the proof supplied by B. Jawerth in [31] for the case $w \in A_1$. This is enough, for instance, for the application given in Theorem 4.7.

Proof for $w \in A_1$. We use inequality (1.16) and the fact that $w \in A_1$ to write

$$\begin{aligned} w(\{x \in Q : w(x) > t\}) &\leq 2^n t |\{x \in Q : Mw(x) > t\} \\ &\leq 2^n t |\{x \in Q : w(x) > [w]_{A_1}^{-1} t\}|, \end{aligned}$$

for $t > w_Q$. Using (A.1) we have

$$\begin{aligned} \int_Q w^r &= (r-1) \int_0^\infty t^{r-2} w(\{x \in Q : w(x) > t\}) dt \\ &\leq w_Q^{r-1} w(Q) + (r-1) 2^n \int_{w_Q}^\infty t^{r-1} |\{x \in Q : w(x) > [w]_{A_1}^{-1} t\}| dt \\ &\leq \frac{w(Q)^r}{|Q|^{r-1}} + \frac{r-1}{r} 2^n [w]_{A_1}^r \int_Q w^r. \end{aligned}$$

Choosing r close enough to 1 the last term can be absorbed by the left-hand side and (4.1) follows. □

Remark 4.2. Sometimes it is convenient to find a range of values of r for which a uniform constant can be taken in the right-hand side of (4.1). The reader can check from the given proof that the interval $[1, 1 + (2^{n+2}[w]_{A_1})^{-1}]$ meets the requirement.

The proof of the general reverse Hölder inequality we offer here is due to Alberto de la Torre. Unpublished by the author, it has been reproduced by some other authors (see, for instance, the appendix in [41]). We need the following lemma.

Lemma 4.3. *If $w \in A_p$ and $0 < s < 1$, then*

$$\frac{1}{|Q|} \int_Q w \leq C \exp\left(\frac{1}{|Q|} \int_Q \log w \right) \leq C \left(\frac{1}{|Q|} \int_Q w^s \right)^{1/s}. \tag{4.2}$$

The same constant can be taken in both terms. In particular, it does not depend on s.

Note that the inequality taking the first and the last term is itself a reverse Hölder inequality with the L^1 norm at the left-hand side.

Proof. If $w \in A_p$, property (ii) of Theorem 2.1 says that for every $q > p$ we have

$$\sup_Q \frac{w(Q)}{|Q|} \left(\frac{1}{|Q|} \int_Q w^{-\frac{1}{q-1}} \right)^{q-1} \leq [w]_{A_p}.$$

Letting q tend to infinity and using Lemma A.2 we obtain

$$\sup_Q \frac{w(Q)}{|Q|} \exp\left(\frac{1}{|Q|}\int_Q \log w^{-1}\right) \leq [w]_{A_p}.$$

This gives the first inequality of (4.2). The second one is immediate from Jensen's inequality (see again Lemma A.2). □

Remark 4.4. Let us denote the best constant in (4.2) as $[w]_{A_{\exp}}$. This constant has been denoted as $[w]_{A_\infty}$ sometimes (see [24, Chapter 9], for instance), but in recent papers the notation $[w]_{A_\infty}$ appears associated to a different constant. Thus we prefer to distinguish it with the special notation $[w]_{A_{\exp}}$. It has been used to improve some of the bounds involving $[w]_{A_p}$. Note that the proof of the lemma shows that $[w]_{A_{\exp}} \leq [w]_{A_p}$. For instance, the bound obtained in Theorem 1.12 can be improved to

$$\|Mf\|_{p,w} \leq C_p([w]_{A_p}[\sigma]_{A_{\exp}})^{1/p}\|f\|_{p,w}.$$

See the paper [30] by T. Hytönen and C. Pérez for results involving mixed $A_p - A_\infty$ constants.

Exercise. Prove that $w \in A_p$ if and only if w and σ satisfy the first inequality in (4.2) and that

$$[w]_{A_p} \leq [w]_{A_{\exp}}[\sigma]_{A_{\exp}}^{p-1}.$$

Proof of Theorem 4.1. Observe first that for $x \in Q$, the value of $M(w\chi_Q)(x)$ only needs to take into account the averages on cubes contained in Q. Then (4.2) implies

$$M(w\chi_Q)(x) \leq [w]_{A_{\exp}} M(w^s\chi_Q)(x)^{1/s}. \tag{4.3}$$

Thus for $t > w_Q$ we have

$$\begin{aligned} w(\{x \in Q : w(x) > t\}) &\leq 2^n t|\{x \in Q : Mw(x) > t\}| \\ &\leq 2^n t|\{x \in Q : [w]_{A_{\exp}} M(w^s\chi_Q)(x)^{1/s} > t\}| \\ &\leq C2^n t^{1-s}[w]_{A_{\exp}}^s w^s(\{x \in Q : w(x)^s > t^s/2\}), \end{aligned}$$

where we use (1.16) in the first inequality, (4.3) in the second and (1.3) in the third one.

We proceed as in the proof of the case $w \in A_1$ with this new estimate to obtain

$$\begin{aligned}\int_Q w^r &= (r-1)\int_0^\infty t^{r-2} w(\{x \in Q : w(x) > t\})dt \\ &\leq w_Q^{r-1} w(Q) + (r-1)2^n C[w]^s_{A_{\exp}} \int_{w_Q}^\infty t^{r-1-s} w^s(\{x \in Q : w(x)^s > t^s/2\})dt \\ &\leq \frac{w(Q)^r}{|Q|^{r-1}} + C_{n,s}\frac{r-1}{\frac{r}{s}-1}\int_Q w^r.\end{aligned}$$

Fix $s < 1$ and choose r close to 1 to absorb the last term by the left-hand side. □

Theorem 4.5. *Let $w \in A_p$.*

(i) If $p > 1$, there exists $\epsilon > 0$ such that $w \in A_{p-\epsilon}$.

(ii) If $p \geq 1$, there exists $\epsilon > 0$ such that $w^{1+\epsilon} \in A_p$.

The details of the proof are left as exercise. For (i) the reverse Hölder inequality is used for $w^{1-p'} \in A_{p'}$, and for (ii) it is used for both w and $w^{1-p'}$. Moreover, from the range in which the reverse Hölder inequality is valid one can estimate the size of ϵ in terms of the A_p constant of the weight.

4.2 Gehring's lemma

The reverse Hölder inequality appeared in a paper of F. Gehring ([22]). Actually, the result known as Gehring's lemma is an improvement of the exponent of a reverse Hölder inequality. Here is the statement.

Lemma 4.6. *If w satisfies the reverse Hölder inequality (4.1) with exponent r for all cubes, then there exists $\delta > 0$ such that w satisfies a reverse Hölder inequality with exponent $r + \delta$.*

Apart from the original paper, the proof of the lemma can be found in [21], for instance. A different proof based on Hardy's operator and rearrangement due to M. Milman is in [35].

Note that Gehring's lemma together with the reverse Hölder inequality for w^s of Lemma 4.3 gives Theorem 4.1 at once. Alternatively, the proof given above for A_p weights is also a proof of Gehring's lemma. Indeed, saying that

w satisfies the reverse Hölder inequality (4.1) with exponent r is the same as saying that $u = w^r$ satisfies

$$\frac{1}{|Q|}\int_Q u \leq C\left(\frac{1}{|Q|}\int_Q u^{1/r}\right)^r.$$

This is all we needed to show that u satisfies a reverse Hölder inequality (with the L^1-norm at the right-hand side), which in turn is an improved reverse Hölder inequality for w.

4.3 Characterization of A_1 weights

Reverse Hölder inequality together with Theorem 2.2 for A_1 gives a characterization of the A_1 class.

Theorem 4.7. *The weight w is in A_1 if and only if there exist a locally integrable function f such that $Mf(x) < +\infty$ a. e., $\delta \in [0,1)$, and a bounded function b with b^{-1} also bounded, such that*

$$w(x) = b(x)Mf(x)^\delta \quad a.e.$$

The if part is a consequence of Theorem 2.2. For the only if part, if r is the exponent in (4.1) for w, take $f = w^r$ and $\delta = 1/r$. The details are left to the reader.

4.4 Failure of the reverse Hölder inequality for some bases

Let us consider the basis $\mathcal{B}_0 = \{(0,b) : b > 0\}$ of $(0,\infty)$ of Subsection 1.6.4 with its associated $A_{p,\mathcal{B}_0}$ weights. It is proved in [13] that these classes need not satisfy several properties proved in this chapter for the usual A_p weights:

- There exists $w \in A_{p,\mathcal{B}_0}$ such that $w \notin A_{q,\mathcal{B}_0}$ for any $q < p$. An example is $w = u^{1-p}$ for

$$u(x) = \chi_{\Omega^c}(x) + \sum_{k=1}^{\infty} \frac{1}{(x-2^k)^2}\chi_{I_k}(x),$$

  where $I_k = (2^k + 2^{-k}, 2^k + 1)$ and $\Omega = \bigcup_{k=1}^{\infty} I_k$.

- This u is in $A_{p,\mathcal{B}_0}$ for every $p \in (1,\infty)$, but it does not satisfy a reverse Hölder inequality on intervals of the form $(0,b)$.

On the other hand, a weight can satisfy a reverse Hölder inequality over the sets in $\mathcal{B}_0$ and vanish in a set of positive measure. An example is $w = \chi_{(0,1)\cup(2,\infty)}$. This weight does not belong to any $A_{p,\mathcal{B}_0}$.

4.5 RH_r classes

For a given $r > 1$ we define the class RH_r as the class of weights satisfying (4.1) for the precise exponent r. Unlike the A_p classes, RH_r classes are decreasing: $RH_r \subset RH_s$ for $1 < s < r < \infty$. The structure of the RH_r classes has been studied, for instance, in [8].

For $w \in RH_r$, let $[w]_{RH_r}$ be the best constant in (4.1) for w.

Exercise. Prove that $w \in A_p \cap RH_r$ if and only if $w^r \in A_{r(p-1)+1}$. Moreover,

$$[w^r]^{1/r}_{A_{r(p-1)+1}} \leq [w]_{A_p}[w]_{RH_r},$$

and

$$[w]_{A_p}, [w]_{RH_r} \leq [w^r]^{1/r}_{A_{r(p-1)+1}}.$$

Exercise. Prove that if $w \in A_p$ for some p and $w^r \in A_1$ for some $r > 0$, then $w \in A_1$. Moreover,

$$[w]_{A_1} \leq [w]_{A_p}[w^r]^{1/r}_{A_1}.$$

Chapter 5

A_∞ weights

Since the union of the A_q classes for $q < p$ is the same as A_p, a natural name for the union of all the A_p classes is A_∞. Characterizations of A_∞ are necessary and sufficient conditions for being an A_p weight, without specifying a value of p. There are many such characterizations and in the literature often one of them appears as the definition of A_∞, and the fact that it is equivalent to being in the union of the A_p classes is given as a theorem.

The first appearences of the A_∞ classes in the literature are due to B. Muckenhoupt in [37] and to R. R. Coifman and C. Fefferman in [4]. Muckenhoupt defines A_∞ as formed by the weights w satisfying the following condition: given $\epsilon > 0$ there exists $\delta > 0$ such that if Q is a cube, $E \subset Q$ and $|E| < \delta|Q|$, then $w(E) < \epsilon w(Q)$. The definition chosen by Coifman and Fefferman is as follows: there are constants $C, \delta > 0$ such that for any cube Q and any measurable $E \subset Q$ it holds that

$$\frac{w(E)}{w(Q)} \leq C \left(\frac{|E|}{|Q|} \right)^{\delta}. \tag{5.1}$$

They also proved that this condition is equivalent to being in A_p for some $p > 1$.

Other characterizations of A_∞ were given by N. Fujii in [18], an apparently less-known paper. Later, S. Hruščev in [28] and independently J. García-Cuerva and J. L. Rubio de Francia in the book [20] proved that the reverse Jensen inequality of Lemma 4.3 (the first inequality in (4.2)) also implies that w is in A_∞. All the mentioned characterizations including those in Fujii's paper were also collected in the survey [15].

The next theorem gives the equivalence of several characterizations.

Theorem 5.1. *Let w be a locally integrable nonnegative function in $\mathbf{R}^n$. The following are equivalent.*

(1) $w \in \bigcup_{1<p<\infty} A_p$.

(2) *There exists $C > 0$ such that for every cube it holds that*

$$\frac{1}{|Q|}\int_Q w \leq C \exp\left(\frac{1}{|Q|}\int_Q \log w\right).$$

(3) $w \in \bigcup_{1<r<\infty} RH_r$.

(4) *There exists $C > 0$ such that for every cube it holds that*

$$\int_Q M(w\chi_Q) \leq Cw(Q).$$

(5) *There exists $C > 0$ such that for every cube it holds that*

$$\int_Q w \log^+ \frac{w}{w_Q} d\mu \leq Cw(Q).$$

(6) *There exist $\alpha, \beta \in (0,1)$ such that for every cube Q and every measurable set E contained in Q for which $|E| < \alpha|Q|$ it holds that $w(E) \leq \beta w(Q)$.*

(6') *There exist $\alpha, \beta \in (0,1)$ such that for every cube Q it holds that*

$$w(\{x \in Q : w(x) \geq \frac{w_Q}{\alpha}\}) \leq \beta w(Q).$$

(7) *There exist $C > 0$ and $\alpha < 1$ such that for every cube Q and $t > w_Q^{-1}$ it holds that*

$$|\{x \in Q : w(x) \leq t^{-1}\}| \leq Ctw(\{x \in Q : w(x) \leq (\alpha t)^{-1}\}).$$

There are more conditions equivalent to being an A_∞ weight. In particular, (6), (6') and (7) have equivalent symmetric expressions with the roles of the Lebesgue measure and the measure $w(x)dx$ interchanged. Also Coifman-Fefferman condition (5.1) has been removed from the list. We recover some of the missing characterizations in Theorem 5.2 below. The order of the conditions in the theorem has been adapted to the chain of implications used in the proof, but there are other ways to arrange the proof. See, for instance, Appendix II in [39], and also [38], where the weights are given in a more general setting, but with similar characterizations.

Condition (4) and (5) appeared in Fujii's paper [18][1]. Condition (4) was later rediscovered and used by J. M. Wilson (in [48], for instance). The best constant in (4) has been defined as $[w]_{A_\infty}$ in recent works dealing with mixed $A_p - A_\infty$ bounds for several operators. It has been preferred to $[w]_{A_{\exp}}$ of Remark 4.4, which is bigger.

Proof of Theorem 5.1. The implications (1) $\Rightarrow$ (2) $\Rightarrow$ (3) have been proved in the previous chapter.

(3) $\Rightarrow$ (4): Assume that w satisfies (4.1) (with exponent r). Then

$$\int_Q M(w\chi_Q) \le \left(\int_Q M(w\chi_Q)^r\right)^{1/r} |Q|^{1/r'} \le C\left(\int_Q w^r\right)^{1/r} |Q|^{1/r'} \le Cw(Q),$$

using Hölder's inequality, the boundedness of M in L^r (unweighted) and (4.1).

(4) $\Rightarrow$ (5): Integrating inequality (1.16),

$$\frac{1}{t}\int_{\{x\in Q: w(x)>t\}} w \le 2^n |\{x \in Q : M(w\chi_Q)(x) > t\}|,$$

for $t > w_Q$ we get

$$\begin{aligned}\int_{w_Q}^\infty \frac{1}{t}\int_{\{x\in Q: w(x)>t\}} w(x)dx\, dt &\le 2^n \int_{w_Q}^\infty |\{x \in Q : M(w\chi_Q)(x) > t\}|\, dt \\ &\le 2^n \int_Q M(w\chi_Q).\end{aligned}$$

[1] Actually in Fujii's version of (4) the integral of the left-hand side is over $2Q$ instead of Q, but it is not hard to see that the integral of $M(w\chi_Q)$ over $2Q \setminus Q$ is controlled by the integral over Q. Another characterization of A_∞ given by Fujii replaces M with a Riesz transform.

The first term gives the left-hand side of (5) and (4) gives the right-hand side.

(5) $\Rightarrow$ (6): Let $E \subset Q$ such that $|E| < \alpha|Q|$. The inequality $ab \leq a\log a - a + e^b$ holds for $a > 1$ and $b > 0$. Assume $w_Q = 1$. We have

$$\begin{aligned} w(E) &= \int_{E\cap\{w\leq 1\}} w + \int_{E\cap\{w>1\}} w \leq |E| + \frac{1}{b+1}\int_E (w\log^+ w + e^b) \\ &\leq \left(1 + \frac{e^b}{b+1}\right)|E| + \frac{C}{b+1}|Q|, \end{aligned}$$

where C is the constant of (5). With $b = 2C - 1$, choose α such that $\alpha\left(1 + e^b/(b+1)\right) < 1/4$. As $w(Q) = |Q|$, (6) holds for such α and $\beta = 3/4$.

(6) $\Rightarrow$ (6'): Let $E = \{x \in Q : w(x) \geq w_Q/\alpha\}$. Then

$$|E| = \int_E w^{-1}w \leq \frac{\alpha}{w_Q}w(E) \leq \alpha|Q|.$$

Therefore (6') follows at once.

(6') $\Rightarrow$ (6): Let $E \subset Q$. Then

$$\begin{aligned} w(E) &\leq w(\{x \in E : w(x) \geq \frac{w_Q}{\alpha}\}) + w(\{x \in E : w(x) < \frac{w_Q}{\alpha}\}) \\ &\leq \beta w(Q) + \frac{w_Q}{\alpha}|E| \leq \left(\beta + \frac{1}{\alpha}\frac{|E|}{|Q|}\right)w(Q). \end{aligned}$$

Choose α' such that $\beta' = \beta + \alpha'/\alpha < 1$. Then (6) holds with α' and β' instead of α and β.

(6) $\Rightarrow$ (7): The measure $w(x)dx$ is doubling due to (6)[2]. Then we can apply the results mentioned in Remark 1.16 taking as f the function w^{-1}. For each $t > |Q|/w(Q)$ we obtain a sequence $\{Q_j\}$ of dyadic cubes in Q for which

$$t < \frac{|Q_j|}{w(Q_j)} < Ct.$$

Moreover, the set $\{x \in Q : w(x) \leq t^{-1}\}$ is contained (up to a set of measure zero) in the union of the cubes Q_j, because for almost every x outside this union we have $w^{-1}(x) \leq t$.

[2]The doubling property is better seen if (6) is written in the form: $|E| \geq (1-\alpha)|Q|$ implies $w(E) \geq (1-\beta)w(Q)$. This is deduced applying (6) to $Q \setminus E$ instead of E.

We deduce that

$$|\{x \in Q : w(x) \le t^{-1}\}| \le \sum_j |Q_j| \le Ct \sum_j w(Q_j).$$

Using (6') we have that

$$w(Q_j) \le w(\{x \in Q_j : w(x) < \frac{w_{Q_j}}{\alpha}\}) + \beta w(Q_j).$$

Since $w_{Q_j} < t^{-1}$ by construction, we obtain

$$w(Q_j) \le \frac{1}{1-\beta} w(\{x \in Q_j : w(x) < (\alpha t)^{-1}\}).$$

Summing in j we obtain (7).

(7) $\Rightarrow$ (1): Let $\epsilon > 0$. We have

$$\begin{aligned}
\int_Q w^{-\epsilon} &\le \epsilon \int_0^\infty t^{\epsilon-1} |\{x \in Q : w(x)^{-1} > t\}|\, dt \\
&\le \left(\frac{|Q|}{w(Q)}\right)^\epsilon |Q| + \epsilon \int_{w_Q^{-1}}^\infty t^\epsilon w(\{x \in Q : w(x)^{-1} > \alpha t\})\, dt \\
&\le \left(\frac{|Q|}{w(Q)}\right)^\epsilon |Q| + \frac{\epsilon}{\alpha^{1+\epsilon}(1+\epsilon)} \int_Q w^{-(1+\epsilon)} w.
\end{aligned}$$

Choosing ϵ small enough the last term is absorbed by the first one and we obtain

$$\left(\frac{1}{|Q|}\int_Q w^{-\epsilon}\right)^{1/\epsilon} \frac{w(Q)}{|Q|} \le C.$$

This is the A_p condition for $p = 1 + 1/\epsilon$. □

Note that the last step of the proof can be read as saying that w^{-1} satisfies a reverse Hölder inequality of exponent $1+\epsilon$ with respect to the measure $w(x)dx$. Therefore, the A_p condition itself is a form of reverse Hölder inequality.

Theorem 5.2. *Let w be a locally integrable nonnegative function in $\mathbf{R}^n$. The following conditions are also equivalent to all those appearing in Theorem 5.1:*

(8) *There exist $\delta, C > 0$ such that (5.1) holds for every cube Q and every measurable set E contained in Q.*

(9) *There exist $\alpha, \beta \in (0,1)$ such that for every cube Q it holds that*

$$|\{x \in Q : w(x) \le \alpha w_Q\}| \le \beta |Q|.$$

(10) *There exist $C > 0$ and $\beta < 1$ such that for every cube Q and $t > w_Q$ it holds that*

$$w(\{x \in Q : w(x) > t\}| \le Ct|\{x \in Q : w(x) > \beta t\}|.$$

Proof. (3) $\Rightarrow$ (8): Using Hölder's inequality and (4.1) we have

$$w(E) = \int_E w \le \left(\int_Q w^r \right)^{1/r} |E|^{1/r'} \le C \left(\frac{|E|}{|Q|} \right)^{1/r'}.$$

(8) $\Rightarrow$ (3): We can assume $w_Q = 1$, that is, $w(Q) = |Q|$ (multiplying by a constant if needed). Let $E_t = \{x \in Q : w(x) > t\}$. From (3') we deduce

$$t|E_t| \le w(E_t) \le C|E_t|^\delta |Q|^{1-\delta},$$

which gives $|E_t| \le Ct^{1/(1-\delta)}|Q|$. Then

$$\int_Q w^r = r \int_0^\infty t^{r-1} |E_t| dt \le |Q| + Cr \int_1^\infty t^{r-1-1/(1-\delta)} |Q| dt.$$

Take $r < 1/(1-\delta)$ and the reverse Hölder inequality follows.

Condition (6) in Theorem 5.1 can be reversed: $w(E) < (1-\beta)w(Q)$ implies $|E| < (1-\beta)|Q|$. Then one can follow the proof of Theorem 5.1 interchanging the roles of $w(x)dx$ and Lebesgue measure to deduce (9) and (10) from (6). From (10) we obtain a reverse Hölder inequality for w. We leave the details of this part to the reader. □

Gehring's lemma 4.6 says that if w satisfies a reverse Hölder inequality with exponent r, this exponent can be further improved to some $r + \delta$. The equivalence between conditions (3) and (5) in Theorem 5.1 can be read in similar terms saying that from the $L \log L$ integrability at the left-hand side of the inequality we jump to an L^r-integrability. The following new characterization taken from [14] shows than even $L \log L$ can be weakened.

Theorem 5.3. *Let w be locally integrable and nonnegative. The following condition is also equivalent to all those given in Theorems 5.1 and 5.2:*

(11) *Let ϕ be a nondecreasing function defined on $(1,+\infty)$ with values in $[0,+\infty)$ such that $\lim_{t\to+\infty}\phi(t)=+\infty$ and*

$$\lim_{t\to+\infty}\frac{\phi(t)}{t^{r-1}}=0 \quad \textit{for all } r>1.$$

There exists $C>0$ such that for every cube Q it holds that

$$\int_{\{x\in Q:w(x)>w_Q\}} w(x)\phi\left(\frac{w(x)}{w_Q}\right)dx \le Cw(Q). \tag{5.2}$$

Proof. (3) $\Rightarrow$ (11): Assume that w satisfies (4.1) with exponent r. Using that $\phi(t)\le Ct^{r-1}$ we have

$$\int_{\{x\in Q:w(x)>w_Q\}} w(x)\phi\left(\frac{w(x)}{w_Q}\right)dx \le C\int_Q \frac{w^r}{w_Q^{r-1}} \le Cw(Q).$$

(11) $\Rightarrow$ (6): From the assumption of this theorem we prove property (6) of Theorem 5.1. For each $s>\phi(1)$ define $\phi^{-1}(s)=\sup\{t:\phi(t)\le s\}$.

Given $E\subset Q$, let $E_1=\{x\in E: w(x)\le \phi^{-1}(2C)w_Q\}$, where C is the constant in (5.2), and $E_2=E\setminus E_1$. We have the estimates

$$\int_{E_1} w \le \phi^{-1}(2C)w_Q|E|$$

and

$$\int_{E_2} w \le \frac{1}{2C}\int_{\{x\in Q:w(x)>w_Q\}} w(x)\phi\left(\frac{w(x)}{w_Q}\right)dx \le \frac{1}{2}w(Q).$$

Adding both estimates we get

$$w(E)\le\left(\phi^{-1}(2C)\frac{|E|}{|Q|}+\frac{1}{2}\right)w(Q).$$

Choosing α such that $\alpha\phi^{-1}(2C)<1/4$, (6) holds with $\beta=3/4$. □

Actually we have different characterizations for different functions ϕ. The choice $\phi(t)=\log t$ gives the previous condition (5) of Theorem 5.1. But we can clearly take ϕ with much smaller growth.

Remark 5.4. The characterizations of A_∞ given in this chapter are valid for the usual A_p classes, but not necessarily for the weights associated to general bases in the sense of Section 1.6. This question is addressed in [14].

We already mentioned in Section 4.4 that conditions (1) and (3) of Theorem 5.1 for the basis $\mathcal{B}_0 = \{(0, b) : b > 0\}$ of $(0, \infty)$ are independent. Another result in [13] shows that for the same basis there are weights satisfying (2) but not (1). An example is

$$w(x) = \chi_{\Omega^c}(x) + \sum_{k=1}^{\infty} (x - 2^k)^k \chi_{I_k}(x),$$

where $I_k = (2^k, 2^k + 1)$ and $\Omega = \cup_{k=1}^{\infty} I_k$.

Appendix A

Toolbox

A.1 Distribution function and L^p norms

Let (X, μ) be a measure space and let $f : X \to \mathbf{C}$ be a measurable function. The function defined for $t \in (0, +\infty)$ by $\mu(\{x \in X : |f(x)| > t\})$ is the *distribution function* of f (associated with μ). We can use it to represent the $L^p(\mu)$ norm of a function.

Lemma A.1. *Let $\phi : [0, \infty) \to [0, \infty)$ be differentiable, increasing and such that $\phi(0) = 0$. Then*

$$\int_X \phi(|f(x)|)\, d\mu = \int_0^\infty \phi'(t)\mu(\{x \in X : |f(x)| > t\})\, dt.$$

To prove this result it is enough to observe that the left-hand side is equivalent to

$$\int_X \int_0^{|f(x)|} \phi'(t)\, dt\, d\mu$$

and then change the order of integration. If, in particular, $\phi(t) = t^p$ for $p > 0$, then

$$\int_X |f(x)|^p\, d\mu = p \int_0^\infty t^{p-1}\mu(\{x \in X : |f(x)| > t\})\, dt. \tag{A.1}$$

Weak-type inequalities for an operator T are bounds for the distribution function of Tf in terms of the norm of f. The inequality

$$\mu(\{x : |f(x)| > t\}) \leq \int_{\{x:|f(x)|>t\}} \frac{|f(x)|^p}{t^p}\, d\mu(x),$$

known as Chebyshev's inequality, shows that strong-type inequalities imply weak-type inequalities.

A.2 Jensen's inequality

Lemma A.2. *Let μ be a probability measure on X (that is, $\mu(X) = 1$) and f a nonnegative integrable function on X. The function*

$$h(s) = \left(\int_X f^s d\mu \right)^{1/s}$$

is increasing in $(0, \infty)$, and

$$\lim_{s \to 0+} h(s) = \exp\left(\int_X \log f d\mu \right),$$

where $e^{-\infty}$ is interpreted as 0.

The fact that h is increasing is a consequence of Hölder's inequality. The interesting part is the value of the limit when s tends to 0. A hint for proving it is in Exercise 1.1.3 of [23]. The lemma implies

$$\exp\left(\int_X \log f d\mu \right) \leq \left(\int_X f^s d\mu \right)^{1/s},$$

which is (one form of) Jensen's inequality.

A.3 Covering lemmas

In the one-dimensional case there is an easy covering lemma.

Lemma A.3. *Let $\{I_k : k = 1, \ldots, N\}$ be a finite family of intervals in $\mathbf{R}$. There exists a subfamily whose union is the same as the union of the original one and such that no point is in more than two intervals of the selected subfamily.*

The result is not true in higher dimensions, not even with a constant depending on the dimension. Instead, the following *Vitali-type covering lemma* is enough for many results. A form of this lemma was used by Wiener in [47] to deal with the weak-type $(1, 1)$ inequality for the maximal operator.

Lemma A.4. *Let $\{Q_\alpha : \alpha \in \mathcal{A}\}$ be a finite family of cubes in $\mathbf{R}^n$. Then we can extract from it a subfamily $\{Q_j : j = 1, \dots, N\}$ of disjoint cubes such that*

$$\bigcup_{\alpha \in \mathcal{A}} Q_\alpha \subset \bigcup_{j=1}^{N} 3Q_j.$$

The reader can try to prove both lemmas. For the second one the iterative selection process consists in taking the largest possible cube not intersecting those previously selected.

The family obtained in Lemma A.4 is formed by disjoint cubes, but we need to dilate them to cover the original family. *Besicovitch covering lemma* is closer to the one-dimensional Lemma A.3, but it needs a special assumption on the starting family of cubes.

Lemma A.5. *There is a constant C_n depending only on n with the following property. Let E be a bounded subset of $\mathbf{R}^n$. Assume that for each $x \in E$ we have a cube Q_x centered at x. Then it is possible to select from the cubes Q_x a (possibly finite) family covering E such that each point is at most in C_n cubes of the selected family.*

A proof of Besicovitch covering lemma can be found in [25] or [34].

A.4 Interpolation

We give the statement of the most usual interpolation theorems, Marcinkiewicz and Riesz-Thorin. The second one is applied to linear operators, the first one to sublinear operators. We recall that an operator T is sublinear if

$$\begin{aligned} |T(f_1 + f_2)(x)| &\le |T(f_1)(x)| + |T(f_2)(x)|, \\ |T(\lambda f)(x)| &= |\lambda||T(f)(x)|, \quad \lambda \in \mathbf{C}. \end{aligned}$$

Theorem A.6 (Marcinkiewicz Interpolation)**.** *Let (X, μ) and (Y, ν) be measure spaces. Let $1 \le p_0 < p_1 \le \infty$. Assume that the sublinear operator T defined on $L^{p_0}(X, \mu) + L^{p_1}(X, \mu)$ satisfies the weak-type inequalities*

$$\sup_{t>0} t\, \nu(\{x \in Y : |Tf(x)| > t\})^{1/p_j} \le A_j \|f\|_{p_j, \mu}, \ j = 0, 1.$$

(If $p_1 = \infty$, we assume that T is bounded from $L^\infty(X, \mu)$ to $L^\infty(Y, \nu)$.) Then T is bounded from $L^p(X, \mu)$ to $L^p(Y, \nu)$ for $p_0 < p < p_1$.

The proof can be found in [11], [20] and [23], for instance. The operator norm of T can be given in terms of A_0, A_1, p_0, p_1 and p.

Theorem A.7 (Riesz-Thorin Interpolation)**.** *Let $1 \leq p_0, p_1 \leq \infty$, $1 \leq q_0, q_1 \leq \infty$, and for $0 < \theta < 1$ define p and q by*

$$\frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \qquad \frac{1}{q} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}.$$

Let T be a linear operator defined on $L^{p_0}(\mu) + L^{p_1}(\mu)$ such that

$$\|Tf\|_{q_j,\mu} \leq M_j \|f\|_{p_j,\mu} \quad \text{for } f \in L^{p_j}(\mu),\ j = 0, 1.$$

Then

$$\|Tf\|_{q,\mu} \leq M_0^{1-\theta} M_1^{\theta} \|f\|_{p,\mu} \quad \text{for } f \in L^p(\mu).$$

The proof can be found in [23].

Remark A.8. The linearity requirement of the Riesz-Thorin interpolation theorem can be sometimes bypassed using a "linearization" of the operator. An operator S defined in a Banach space B is *linearizable* if for every $f \in B$ there exists a linear operator T depending on f such that

$$|Tg(x)| \leq |Sg(x)|, \text{ for all } g \in B, \text{ and } |Tf(x)| = |Sf(x)|.$$

If S satisfies the hypotheses of the theorem, so does T by the first inequality. Riesz-Thorin interpolation is applied to the linear operator T, and the conclusion when applied to f, is valid for Sf. Since the involved constants do not depend on f, the result is obtained for S.

In practice we do not even need $|Tf(x)| = |Sf(x)|$, it is enough to have $|Tf(x)| \geq c|Sf(x)|$, with c independent of f.

A.5 Pointwise convergence from weak-type inequalities

Theorem A.9. *Let $\{T_t\}$ be a family of linear operators on $L^p(X, \mu)$ and define*

$$T^* f(x) = \sup_t |T_t f(x)|.$$

If T^* is weak (p,q) then the set

$$\{f \in L^p(X,\mu) : \lim_{t\to t_0} T_t f(x) = f(x) \ \ a.e.\}$$

is closed in $L^p(X,\mu)$.

This theorem is proved in [11] and [24], for instance.

A.6 Kolmogorov's inequality

Lemma A.10. *Let T be an operator satisfying the weak-type inequality*

$$\sup_{t>0} t\,\mu(\{x : |Tf(x)| > t\}) \le A\|f\|_{1,\mu}. \tag{A.2}$$

Let $0 < \delta < 1$, and a set E of finite μ-measure. Then

$$\int_E |Tf|^\delta \, d\mu \le \frac{A^\delta}{1-\delta}\,\mu(E)^{1-\delta}\|f\|_{1,\mu}^\delta. \tag{A.3}$$

Proof. By (A.1) and (A.2),

$$\begin{aligned}\int_E |Tf|^\delta \, d\mu &= \delta \int_0^\infty t^{\delta-1}\mu(\{x \in E : |Tf(x)| > t\})\, dt \\ &\le \delta \int_0^\infty t^{\delta-1} \min\left(\mu(E), \frac{A}{t}\|f\|_{1,\mu}\right) dt \\ &= \delta \int_0^{A\|f\|_{1,\mu}/\mu(E)} t^{\delta-1}\mu(E)\, dt + \delta \int_{A\|f\|_1/\mu(E)}^\infty t^{\delta-2} A\|f\|_{1,\mu}\, dt.\end{aligned}$$

The desired inequality follows immediately. □

Remark A.11. M. Cotlar observed that the converse is also true: if (A.3) holds for some $0 < \delta < 1$, then (A.2) holds. Indeed, if $E_t = \{x : |Tf(x)| > t\}$,

$$\mu(E_t) \le \int_{E_t} \frac{|Tf|^\delta}{t^\delta}\, d\mu,$$

and using (A.3) we get the weak-type inequality. Since we do not know a priori that $\mu(E_t)$ is finite, we can start working with subsets of finite measure.

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